

Llama 3.2

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Demo Environment

Development Boards: Jetson Orin, RDK X5, Raspberry Pi 5 series mainboard

Note: Due to performance limitations, Jetson Orin Nano 4GB, Raspberry Pi 5 2GB/4GB, and RDK X5 4GB versions require a smaller parameter version to run.

Meta Llama 3.2 is a series of advanced open-source large language models (LLMs) developed by the Meta AI department.

1. Model Size

Model	Size
llama3.2:1b	1.3GB
llama3.2:3b	2.0GB

2. Performance

Lightweight instruction-tuned benchmarks

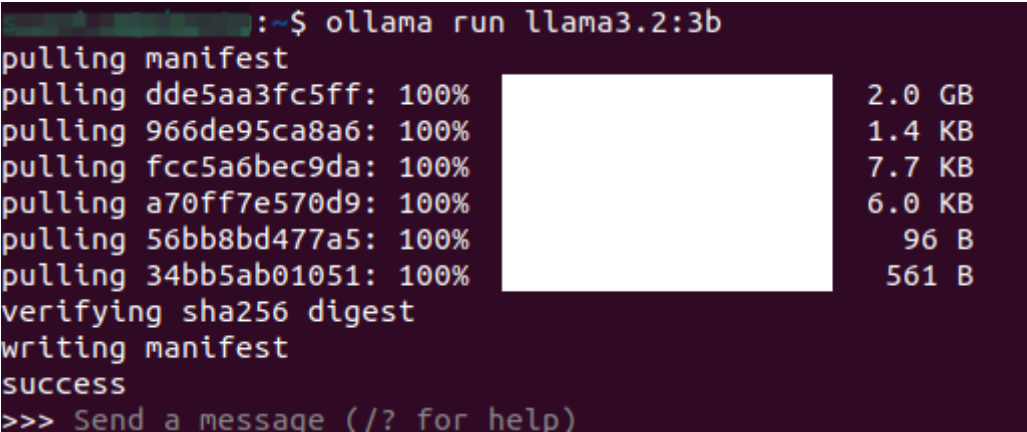
Category Benchmark	Llama 3.2 1B	Llama 3.2 3B	Gemma 2 2B IT (measured)	Phi-3.5-mini IT (measured)
General				
MMLU (5-shot)	49.3	63.4	57.8	69.0
Open-rewrite eval (0-shot, rougeL)	41.6	40.1	31.2	34.5
TLDR9+ (test, 1-shot, rougeL)	16.8	19.0	13.9	12.8
IFEval	59.5	77.4	61.9	59.2
Tool Use				
BFCL V2	25.7	67.0	27.4	58.4
Nexus	13.5	34.3	21.0	26.1
Math				
GSM8K (8-shot, CoT)	44.4	77.7	62.5	86.2
MATH (0-shot, CoT)	30.6	48.0	23.8	44.2
Reasoning				
ARC Challenge (0-shot)	59.4	78.6	76.7	87.4
GPQA (0-shot)	27.2	32.8	27.5	31.9
Hellaswag (0-shot)	41.2	69.8	61.1	81.4
Long Context				
InfiniteBench/En.MC (128k)	38.0	63.3	—	39.2
InfiniteBench/En.QA (128k)	20.3	19.8	—	11.3
NIH/Multi-needle	75.0	84.7	—	52.7
Multilingual				
MGSM (0-shot, CoT)	24.5	58.2	40.2	49.8

3. Using Llama 3.2

3.1 Running Llama 3.2

Use the `run` command to start running the model. If you haven't downloaded this model before, it will automatically pull the model from the Ollama model library:

```
ollama run llama3.2:3b
```



3.2 Starting a Conversation

How many minutes are there in a day?

Response time depends on hardware configuration; please be patient!

```
>>> How many minutes are there in a day?
There are 1440 minutes in a day. This is calculated by multiplying the
number of hours in a day (24) by the number of minutes in an hour (60).

24 hours/day * 60 minutes/hour = 1440 minutes/day

>>> Send a message (/? for help)
```

3.3 Ending a Conversation

Use the shortcut `Ctrl+d` or `/bye` to end the conversation!

3.4 Chinese Dialogue

For those without a Chinese input method, please refer to the Chinese input method switching tutorial.

Chinese Dialogue:

```
root@kali:~/ollama# ollama run llama3.2:3b
>>> 一个小时是多少分钟，请用中文告诉我。
一个小时有60分钟。

>>> Send a message (/? for help)
```

References

Ollama

Official Website: <https://ollama.com/>

GitHub: <https://github.com/ollama/ollama>

Llama 3.2

Official Website: https://www.llama.com/docs/model-cards-and-prompt-formats/llama3_2/

Ollama Corresponding Model: <https://ollama.com/library/llama3.2>